

Lauren Adachi

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ENGINEERING & WORK EXPERIENCE

Amazon

Electrical Hardware Development Engineer II, Just Walk Out Technologies

June 2026-Present

- Lead electrical engineer for Camera program enabling AI-powered [Just Walk Out](#) retail experience.
- Delivered hardware architecture design and cost studies for next generation Camera concepts.

Electrical Hardware Development Engineer II, Amazon Prime Air

May 2025-June 2026

- Served as Responsible Engineer for the Battery Management System (BMS) of 2kW [drone](#) Li-ion battery assembly; owned the full hardware lifecycle including schematic design, PCB layout, DFM, functional test, and EMI qualification.
- Owned sustaining engineering for production Block 20 BMS hardware; led failure analysis (FA) and root cause investigations for field returns and production defects, implementing design changes through the Change Control Board (CCB) to improve fleet-wide BMS reliability and manufacturing yield.
- Led production release of Block 30 BMS hardware, delivering Controller and Analog Front End (AFE) PCBs and Emergency Flight Termination system PCB. Developed next-generation Block 40 BMS for a novel battery pack design, integrating Solid State Circuit Breakers (SSCBs) and HW-based fuse control.
- Architected, designed, and validated a portable battery discharger featuring self-powered architecture that eliminates AC input requirements; integrated MCU-controlled SSCBs and a secondary hardware-redundant relay layer for fail-safe, autonomous operation in remote environments.

Blue Origin

Avionics Design Engineer II, Lunar Permanence Avionics

Dec 2022-May 2025

- Delivered bidirectional Voltage Source Inverter PCBA to drive and control >200V, >100krpm, >2kW motor [for Lunar applications](#). Traded circuit topologies, simulated circuits, performed derating calculations, designed schematic, executed high power PCB layout, delivered & executed test plan, debugged, collected comprehensive functional test data at multiple power levels. Developed Python GUI to command, receive, and log data from FPGA.
 - Received Blue Origin Liftoff Award for exemplary demonstration of Leadership Principles (October 2023)
- Developed flight-like engineering design units (EDUs) for Voltage Source Inverter Card and 3-Phase Disconnect Card.
- Served as Avionics lab manager: developed lab from scratch including specifying & managing installation of equipment (high voltage supplies and e-loads, measurement devices, harnessing), & developed in-house >200V PCB test capability
- Led Single-Event Effects Radiation testing of >30 electrical chips. Designed test PCBs with no hardware failures.

Avionics Systems Engineer, Space Destinations Avionics

Jul 2022 - Dec 2022

- Responsible Engineer for RIO (Remote Input/Output) and Data Acquisition subsystems; owned vehicle electrical interfaces, system architecture, and hardware trade studies for [Space Destinations Avionics](#).
- Engineering Project Manager for New Glenn RIO PCBA; led requirements, radiation/EEE testing, and obsolescence mitigation from part selection through procurement.

Brown Space Engineering | *Co-president and Avionics Hardware Division Leader*

Aug 2018 – May 2022

- Led team of 30 undergraduates to design [CubeSat](#) by defining and managing engineering requirements and goals, liaising with technical reviewers and advisors, and ensuring cohesive integration between satellite subsystems
- Awarded [\\$300,000 in launch services](#) from NASA's ELaN satellite educational launch program. Wrote technical proposal detail and electronics systems overview and managed power and link budgets for satellite
- Led Avionics Hardware subgroup of 10 undergraduates to create high-level design and prototyping of electronics systems including block diagrams, KiCAD schematics, and board designs for power, radio, and control systems

SKILLS

Design & Analysis: Altium, KiCAD, EAGLE, Cadence Virtuoso, LTSpice, ModelSim, MATLAB, Simulink, Circuit derating.

Technical Competencies: Power Electronics, Battery Management Systems (BMS), Inverters, Solid State Circuit Breakers (SSCBs), 50-300V Design, Schematic Capture & Layout, High-Speed and High-Power PCB Layout, DfX (Manufacturing/Test), Digital Communication (I2C, SPI, CAN), EMI/EMC Testing, Radiation Testing (Single-Event Effects), Root Cause Analysis (RCA).

Hardware-in-the-Loop (HIL) & Lab: Oscilloscopes, Spectrum Analyzers, DMMs, Differential Probes, Electronic Loads, High-Voltage Power Supplies, Soldering.

Embedded & Software: Python (GUI & Data Analysis), C (STM32), FreeRTOS, Verilog (FPGA), Java, RISC-V, Git.

EDUCATION & AWARDS

Brown University | Electrical Engineering B.S. | GPA: 3.9/4.0

Aug 2018-May 2022

Brooke Owens Fellow, Class of 2021